

CASE STUDY

Fairness Intervention for Disproportionate Rejections of South Asian Women in IVRAS

Why This Case Study Was Selected

This case study uses the Immigration Visa Risk Assessment System (IVRAS) to evaluate the Fairness Implementation Playbook.

IVRAS was selected because it is a high-risk AI system that directly impacts individuals' legal rights, mobility, and access to opportunities. Another dominating factor is that I lived abroad and there were several factors that I could relate to, which made it the most relatable scenario. Unlike other systems, IVRAS involves multiple stages, including OCR, NLP processing, biometric verification, and decision thresholds, making it susceptible to complex, multi-stage bias.

This allows the playbook to be tested under:

- High-risk regulatory conditions
- Intersectional bias scenarios
- Multi-layer intervention requirements

If the playbook performs effectively in this environment, it can be reliably adapted to other domains such as hiring, finance, and healthcare.

Background

The Immigration Visa Risk Assessment System (IVRAS) is an AI-based system used to assign risk scores to visa applicants, determining whether applications receive automatic approval, require manual review, or are flagged for extended investigation. Because these decisions directly affect individuals' mobility, safety, and legal rights, IVRAS is treated as a high-risk AI system.

This case study is based on findings from the IVRAS Fairness Audit, conducted using the *Fairness Audit Playbook for AI Decision Systems*. The audit analyzed historical IVRAS decision data, including approval and rejection outcomes, OCR error rates across document languages, NLP confidence behavior for non-native English text, biometric confidence signals, and feature usage patterns such as travel frequency. As stated in the audit playbook, all data and metric values used are hypothetical and illustrative, intended to demonstrate methodology rather than reflect production data.

The audit identified consistent fairness issues related to OCR errors, NLP penalties, proxy feature usage, and deployment practices, with the most significant harms emerging under intersectional analysis. When nationality and gender were evaluated together, South Asian women were found to experience disproportionately higher rejection rates, despite being otherwise qualified applicants. Based on the audit's conclusion that these disparities reflected systemic bias rather than increased applicant risk, the issue was escalated for remediation using the Fairness Intervention Playbook.

Application of the Fairness Implementation Playbook

The Fairness Implementation Playbook was applied in a structured workflow:

1. Disparity Detection

Intersectional analysis revealed disproportionately high rejection rates for South Asian women.

2. Causal Fairness Analysis

Root causes were identified across multiple stages, including OCR errors, NLP confidence penalties, proxy feature usage, and decision threshold bias.

3. Risk Classification

IVRAS was classified as a high-risk system, requiring comprehensive intervention across all stages and strict governance.

4. Intervention Selection

Based on causal findings, a layered intervention approach was selected, combining pre-processing, in-processing, and post-processing techniques.

5. Implementation

Interventions were applied across data, model, and decision stages.

6. Validation

Fairness improvements and performance trade-offs were evaluated using defined metrics and thresholds.

7. Governance Approval

Deployment was approved only after validation passed and documentation was complete.

8. Monitoring

Post-deployment monitoring was established to detect drift and reintroduce the workflow if bias reappears.

Fairness Interventions Applied

Using the Fairness Intervention Playbook, audit findings were translated into coordinated remediation actions across the system.

Pre-Processing Interventions (Data-Level)

- Improved OCR handling for non-Latin scripts
- Enhanced normalization of document and facial images
- Rebalancing of training data to reduce under-representation of affected groups

Purpose: Reduce measurement errors identified during the audit.

In-Processing Interventions (Model-Level)

- Reduced dependency on proxy features such as travel frequency
- Introduced constraints to limit group-level error disparities
- Addressed NLP confidence penalties during model training

Purpose: Prevent amplification of known data and feature biases.

Post-Processing Interventions (Decision-Level)

- Adjusted decision thresholds in high-uncertainty score ranges
- Introduced reject-option logic, routing borderline cases to manual review instead of automatic rejection

Purpose: Reduce immediate harm while upstream fixes stabilized.

Validation and Monitoring

Validation followed the Validation Framework defined in the Fairness Audit Playbook, including:

- Before-and-after comparison of fairness metrics
- Intersectional group checks
- Monitoring for new or shifted bias after intervention

Approval for deployment was granted only after fairness gaps were reduced without unacceptable degradation in system performance.

Validation acted as a decision gate, ensuring that interventions were not only implemented but also proven effective before deployment.

Intervention Impact Summary (Illustrative)

Before Intervention

- South Asian women experienced significantly higher false rejection rates under intersectional analysis
- OCR and NLP confidence scores were systematically lower for non-Latin scripts
- Borderline cases were automatically rejected without escalation

After Intervention

- False rejection rate disparities were reduced to within approved thresholds
- Borderline cases were routed to manual review instead of automatic rejection
- Overall system accuracy remained within acceptable limits

Trade-offs Observed

- Manual review volume increased modestly
- Slight increase in processing time for affected cases

Audit Findings and Conclusion

The Fairness Audit Playbook highlights that fairness issues often remain hidden under single-attribute analysis and become visible only through intersectional evaluation. Applying this approach to IVRAS showed that South Asian women were disproportionately affected due to multiple interacting bias sources across the pipeline. These included OCR and data bias for non-Latin scripts, NLP penalties for non-native English writing styles, biometric measurement bias leading to higher false non-match rates, proxy feature use such as travel frequency reflecting passport privilege, and deployment bias from uniform decision thresholds.

Together, these factors systematically reduced confidence scores for South Asian women, increasing rejection rates despite applicants being otherwise qualified. This conclusion was based on intersectional segmentation, consistent bias signals across pipeline stages, and fairness metric failures identified in the audit, including Equal Opportunity gaps, false flagging disparities, OCR parity failures, and calibration issues. The audit framework characterizes these patterns as systemic bias, requiring intervention rather than continued monitoring.

Conclusion. This case study demonstrates how findings from the IVRAS Fairness Audit Playbook, based on hypothetical but realistic data, were translated into layered fairness interventions across data, model, and decision stages, reducing disproportionate rejections

faced by South Asian women while maintaining governance, accountability, and system integrity.

Appendix: Application to Hiring Systems

The same playbook can be applied to AI-based hiring systems, such as resume screening or candidate ranking models.

Context

Hiring systems are high-risk due to their impact on employment opportunities and economic outcomes.

Application Mapping

- **Causal Analysis:** Identify bias from gender indicators or university proxies
- **Pre-Processing:** Remove or rebalance biased features
- **In-Processing:** Apply fairness-aware ranking constraints
- **Post-Processing:** Adjust shortlist thresholds
- **Validation:** Compare selection rates across groups

Key Risks

- Gender bias
- Educational bias
- Career gap bias

Adaptation Insight

While the domain differs, the workflow, governance, and validation structure remain unchanged. Only the features and bias sources differ.

This demonstrates that the playbook is adaptable across domains.